

Master Thesis Supervisor Meeting – Authoritative Numbers Cheat Sheet

Mode-I Phase-Field Convergence & UEL Energy Audit · Date: 08 October 2026, 10:00

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1. Authoritative Mechanical Benchmark Numbers (Job Provenance Audited)

Simulation Model	Elements	K_0 [kN/mm]	$F_{\max}$ [kN]	$u(F_{\max})$ [mm]	Status & Extraction Provenance
Fixed Reference Anchor (1398090 / 1409734)	15,192 (CPE4)	137.945520 (Ref Anchor)	0.757778 (Ref Anchor)	0.005857 (Ref Anchor)	Paper-matched fixed benchmark anchor; unconstrained OLS $N = 400$, $R^2 = 0.99999960$, $b = 4.472 \times 10^{-5}$ kN
Corrected ET1 (1.0%) (1409982.mmaste02)	14,483 (14.1kQ + 0.4kT)	137.909558 (−0.0261%)	0.743701 (+0.279% vs fine)	0.005733 (−2.12%)	Preferred corrected adaptive mesh; 64.12% corridor share; reached $u = 7.89 \mu\text{m}$ (98.5% load drop)
Spatial-Fine 8T SMP (1410504.mmaste02)	57,929 (56.3kQ + 1.6kT)	137.840989 (−0.0758%)	0.741633 (0.000% Fine Ref)	0.005717 (−2.39%)	Internal adaptive spatial convergence reference; full horizon $u = 10.0 \mu\text{m}$, 8T SMP ($S_8 = 3.47\times$), Exit=0
Adaptive ET2 (2.0%) (1410357.mmaste02)	6,112 (5.9kQ + 0.2kT)	137.976065 (+0.0221%)	0.756367 (−0.186%)	0.005841 (−0.27%)	Step-2 tolerance sweep; full horizon $u = 10.0 \mu\text{m}$; 41.07% corridor share; 0 cutbacks, Exit=0
Adaptive ET3 (3.0%) (1410358.mmaste02)	5,189 (5.0kQ + 0.2kT)	137.977506 (+0.0232%)	0.759407 (+0.215%)	0.005876 (+0.32%)	Step-2 tolerance sweep; full horizon $u = 10.0 \mu\text{m}$; 34.59% corridor share; 0 cutbacks, Exit=0
Adaptive ET5 (5.0%) (1410359.mmaste02)	4,692 (4.5kQ + 0.2kT)	138.009080 (+0.0461%)	0.765400 (+1.006%)	0.005926 (+1.18%)	Step-2 tolerance sweep; full horizon $u = 10.0 \mu\text{m}$; 27.51% corridor share; 0 cutbacks, Exit=0

2. UEL Energy Bookkeeping & Parity Numbers

A. Terminal Global Energy Bookkeeping ($u = 10.0 \mu\text{m}$):

- Fixed Benchmark Anchor (1409734): $W_{\text{ext}} = 2.35933$ mJ, $E_{\text{frac}} = 2.34022$ mJ, $E_{\text{elas}} = 0.00116$ mJ, $\varepsilon_{\text{book}} = \mathbf{0.7607\%}$.
- Corrected ET1 (1409982, $u = 7.89 \mu\text{m}$): $W_{\text{ext}} = 2.26738$ mJ, $E_{\text{frac}} = 2.28547$ mJ, $E_{\text{elas}} = 0.00696$ mJ, $\varepsilon_{\text{book}} = \mathbf{1.1048\%}$.
- Spatial-Fine 58k (1410504): $W_{\text{ext}} = 2.52174$ mJ, $E_{\text{frac}} = 2.38194$ mJ, $E_{\text{elas}} = 0.02818$ mJ, $\varepsilon_{\text{book}} = \mathbf{4.4263\%}$.

B. Non-Invasive Mechanical Parity Verification:

- Subroutine source: f42_mixed_uel.for (SHA-256: CE8D5EDC...).
- Mechanical parity verified: $\max |\Delta F| = \mathbf{0.000000}$ kN (0 Newton perturbation).
- Single-IP Extraction Rule: eliminates $4\times$ overcounting on CPE4 visualizers.

C. Crack Symmetry and Localization Band Metrics:

- Crack Centroid Trajectory: $|y_c - 0.500 \text{ mm}| \leq \mathbf{0.000 \mu\text{m}}$ across tested grids.
- Pre-peak ligament profile L_2 difference: $\leq \mathbf{0.32\%}$ (ET1 vs Fine Reference).
- Transverse damage bandwidth: $w_{0.5} \approx \mathbf{15.0 \mu\text{m}} \approx 2l_0$ ($l_0 = 7.5 \mu\text{m}$).
- ET1 corridor mesh resolution: $h_{\min} = 0.760 \mu\text{m}$, median $h = 2.597 \mu\text{m}$.

D. Discretization-Family Framework:

- $\Delta F_{\max}$ between ET1 and Spatial-Fine Reference is only $+\mathbf{0.279\%}$ (Internal Convergence).
- $\Delta F_{\max}$ between ET1 and Fixed Benchmark Anchor is $-\mathbf{1.856\%}$ (Family Offset).
- Fixed structured meshes are themselves mesh-sensitive (~ 0.7255 kN at 69k FEs).

3. Multi-Quantity Convergence & Epistemological Status Summary

Evaluated Quantity	Observed Variation	Authoritative Status	Epistemological Grounding / Evidence Boundary
Initial Stiffness K_0 (Spatial)	$\Delta K_0 \leq 0.076\%$ across all grids	CONVERGED / STABLE	Stable linear elasticity ($u \leq 1.0 \mu\text{m}$); insensitive to crack-tip damage.
Peak Reaction Force $F_{\max}$ (Adaptive)	$\Delta = +0.279\%$ (ET1 vs 58k)	INTERNALLY CONVERGED	Demonstrates high efficiency and internal spatial convergence with $\sim 75\%$ fewer elements.
Peak Reaction Force $F_{\max}$ (Fixed)	$0.7578 \rightarrow 0.7255$ kN (4.26%)	MESH-SENSITIVE	Fixed structured mesh is not peak-force converged; -1.86% is a family offset.
Crack Centroid Symmetry $y_c(x)$	$ y_c - 0.5 \text{ mm} \leq 1.50 \mu\text{m}$	CONVERGED / STABLE	Centroid offset strictly bounded within single element dimension.
Two-Term Bookkeeping Δ_{book}	$+0.76\%$ (Fixed), $+4.43\%$ (Fine)	CONVERGENCE-CONSISTENT	Algebraic difference $W_{\text{trap}} - (E_{\text{elas}} + E_{\text{frac}})$; not a proven balance law.
Global Energy Identity	Non-commuting staggered steps	GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED	Standard ODBs unpersisted for internal Newton paths; identity remains open.